UNIT - 6

SFU : The SFU stands for Switch fuse unit. they are used for distributing power and protecting electrical devices and cables from damage due to fluctuations.

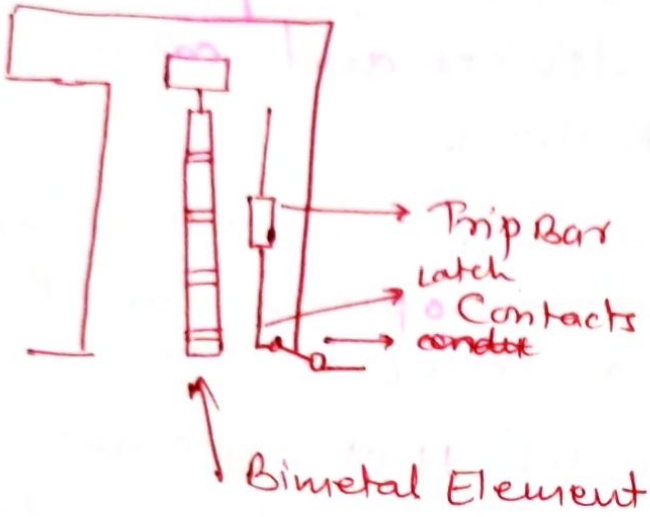
That is a small thin conductor designed to melt and separate into two pieces for the purpose of breaking a circuit breaker is a specially designed switch that automatically opens to ~~crept~~ interrupt circuit current in the event of an overcurrent condition

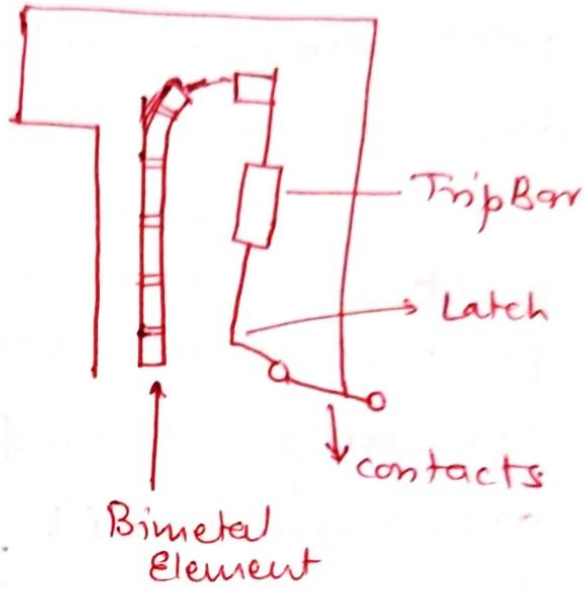
MCB : MCB stands for Miniature circuit breaker. It operates as automatic switch that opens in the event of excessive current flowing the circuit and once the circuit returns to normal, it can be reclosed without any ~~manually~~ manual replacement.

MCB is electromagnetic device that encloses in a molded insulating material.

MCBs are time delay (2.5 milliscond) ~~timing~~ tripping device ~~to which~~ that get operated whenever the overload exist long enough to create a danger to the circuit being protected.

Under normal condition MCB works as a normal switch to make ckt ON and OFF. Under overload conditions, it automatically trips to intruption of circuit.





Classification

Type B MCB: - Trips between 3 and 5 times of full load current
- mainly used for residential applicat

Type C MCB: - Trips between 5 and 10 times full load current.
- Mainly used commercial type of applications where there could be chance of higher value of short ckt current

Type D MCB - Trips between 10 - 20 times full load current.
- Used in specially industrial, commercial where current inrush can be very high. e.g transformer, x-ray machines etc.

ELCB : It stands for earth leakage circuit Breaker. If current leaks from any electrical circuit, there is a insulation failure this must be detected and prevented.

Two Types ① Voltage earth leakage circuit breaker
② Current earth leakage circuit breaker

① Voltage earth Leakage Circuit Breaker.
② Current Earth leakage Circuit Breaker.

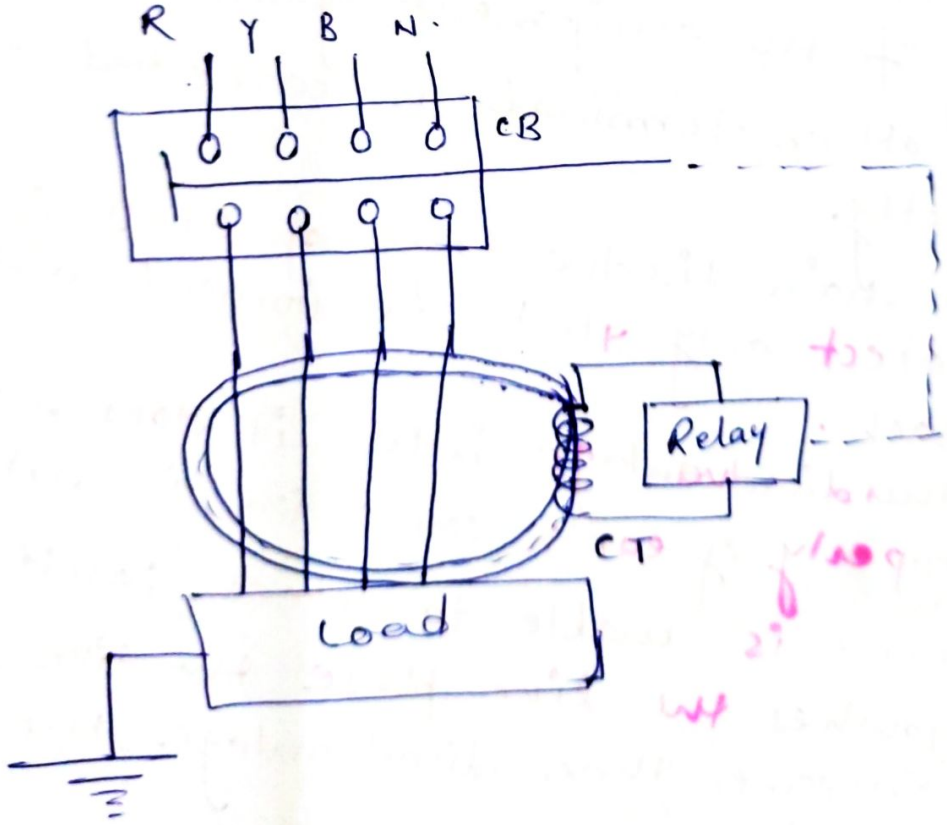
① VoltagELCB : In this device connected between the wires coming from the earth rod to wire attached to the external metalic body of the equipment against earth leakage and other terminal is connected to the earth directly.

- The main disadvantage of ELCB is that it protect only that equipment with which it is attach.
- Other disadvantage is that it does not work properly if earth connection is not proper.
- ELCB is unable to detect fault if someone touches the live phase ~~the live~~ wire.

To overcome these disadvantages the current ELCB is used.

② **Current ELCB :** Current earth leakage circuit breaker is also know as Residual Current Circuit breaker (RCCB). Generally, voltage ELCB is reffered as ELCB and current ELCB is referred as RCCB.

When any earth leakage occurs there is a difference between phase and nutral current. Due to this the resultant mmf will not be zero and this sense by CT (current transformer) and the signal from CT is fed to the circuit breaker (CB) and it open the main power contacts



Earthing

An electrical system is said to be earthed, if the outer surface or non-electrical part is not carring any current and are connected to earth so that the potential is almost zero. The main purpose of earthing is to ensure that all parts of system except live-parts (parts which are carrying currents) are at zero potential

Earthing is essential because:

1. Earthing protects the personal working on it from shock due to short circuit current.
2. It protects the apparatus and personal from high voltage surge and lighting discharge.
3. Earthing ~~protects the ap~~ used to save conducting material & by providing return path.
4. Earth also help in protecting the high rise building from atmospheric lighting

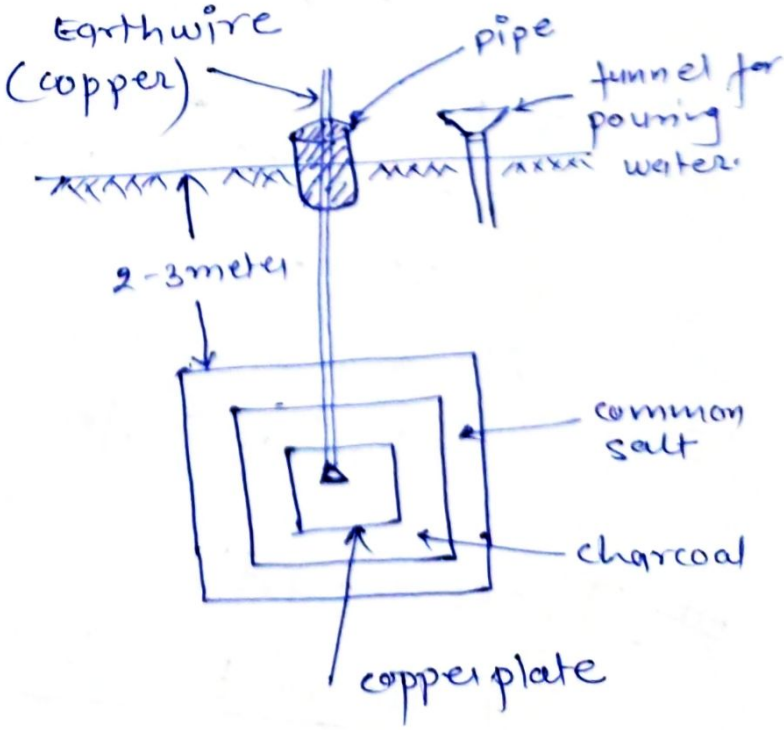
Types of Earth

1. Neutral Earthing : In this the neutral of the system is directly ~~connected~~ connected to the earth
2. Equipment ~~Earthing~~ Earthing : This is the primary type of earthing for home and other buildings.

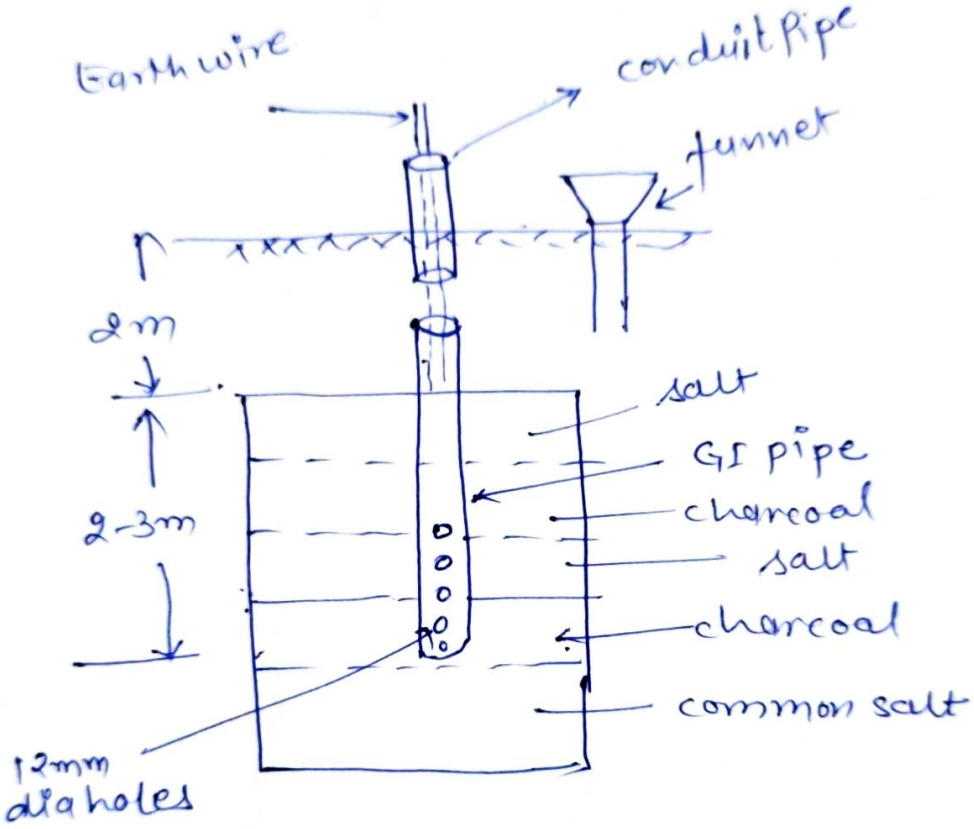
The equipment earthing deals with the safeguard of the non current carring apparatus and metallic conductors.
During fault in apparatus, the earth wire provide the path to the short-circuit current and protect the system from any kind of damage.

Methods of Earthing

① Plate Earthing: In this a plate of galvanised iron is placed inside a deep pit (2 to 3 meter) along with charcoal and sand. The earth wire is connected to earth point of socket and is finally covered.



② **Pipe Earthing:** In this, a pipe of 35-75 mm diameter with a few holes at its lower end is buried in the dig out pit along with sand and charcoal. Pipe is provided with several holes to maintain dampness and maintain conductivity.



③ **Rod Earthing:** It is similar to pipe earthing. In this method uses zinc or copper rods of about 1-1.5 m length and 12-20 mm diameter into the earth

- This method is very quick and cost effective method of earthing.

Molded Case Circuit Breaker (MCCB) : that is

a electromechanical device which protect the circuit from over current and short circuit

- It provides protection by combining a temp. sensitive device with a current sensitive ~~to~~ a ~~fuse device~~. electromagnetic devices act mechanically on the trip mechanism.
- MCCB is an alternative to a fuse device since it does not require replacement once overload is detected and can be easily reset.

Classification

① Thermal trip Unit : During overload the excess current flowing through a piece of bimetal is heated. To trip the CB, this bimetal must deflected enough to physically push the trip bar and disconnect the circuit.

② Electromagnetic Trip Unit : During short ckt the current flowing through the windings increase; which increase the magnetic field strength of the electromagnet and attract the armature. Due to this, the armature rotates the trip bar and circuit breaker trip.

③ Thermal Magnetic trip Unit : Thermal-magnetic circuit breakers use bimetals and electromagnetic assemblies to provide over current protection.

The minor disadvantage is the operating charac. of the breaker, which varies the ambient temp. It is vary usefull where temp. sensitive action is required.

Comparison of MCB and MCCB

- An MCB has less than 100 amps, while an MCCB goes as high as 2500 amps.
- The interrupting rating for an MCB is 18000 amp and up to 200,000 amps for an MCCB.
- MCBs are mostly installed for home use, while an MCCB is generally utilized for commercial or industrial purposes.
- Both are low voltage circuit breakers created to meet standards.
- Some MCCB unit are specially made to respond control signals, usually as standby power.